



Audio Computing Lab

Reverberation and Room Simulation

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Overview

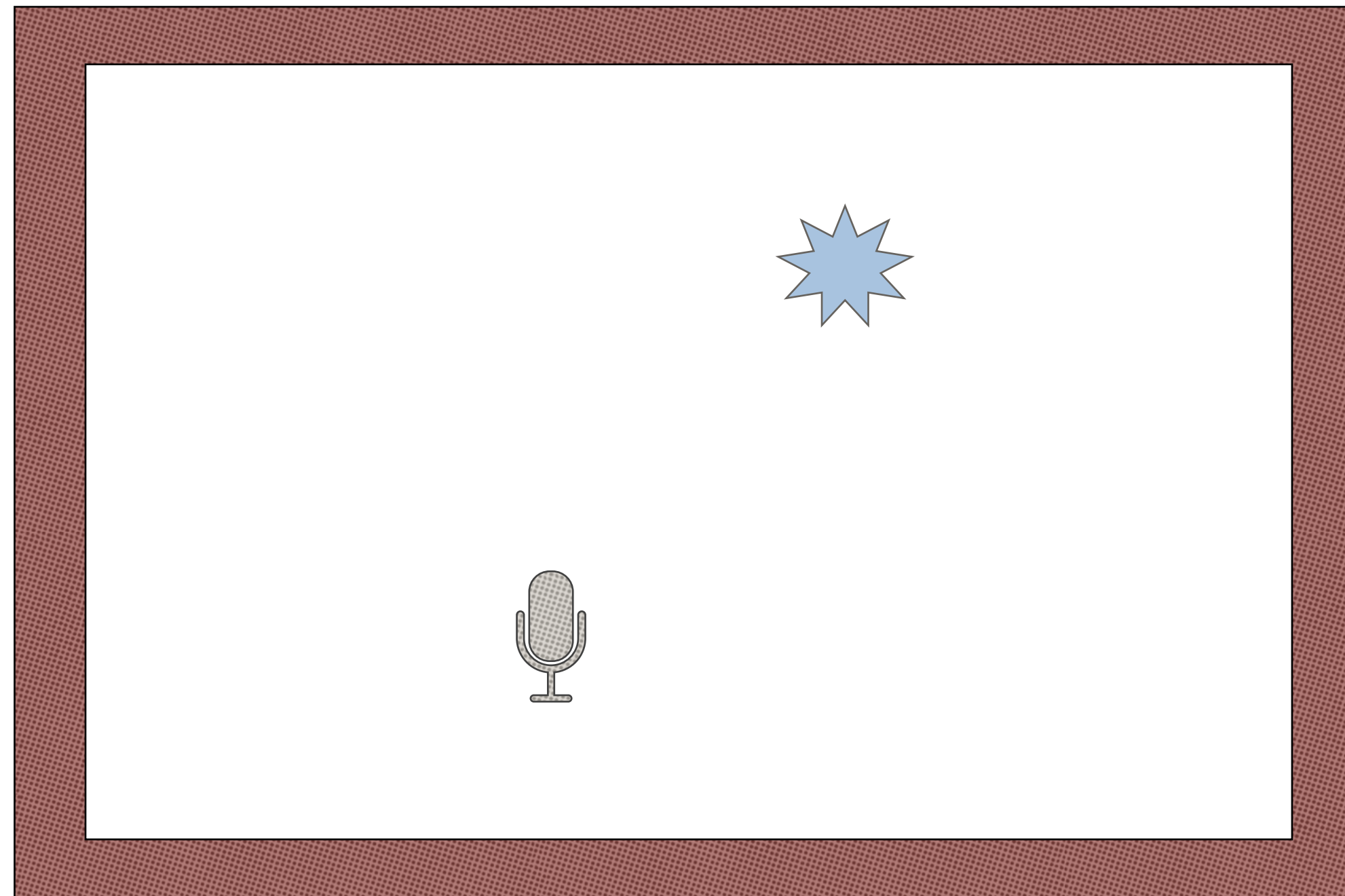
- Digital reverberation
 - Simple filter models
 - Convolution with real impulses
- Measuring reverb from rooms
- Room simulators
 - Image-source model

Reverberation

- Effects that rooms impose on sounds
 - Desirable effect for music, movies, video games
 - Undesirable effect for speech recordings!
- Couple of effects
 - Echoes (repeating the input after a while)
 - Coloring (changing the spectrum of the input)

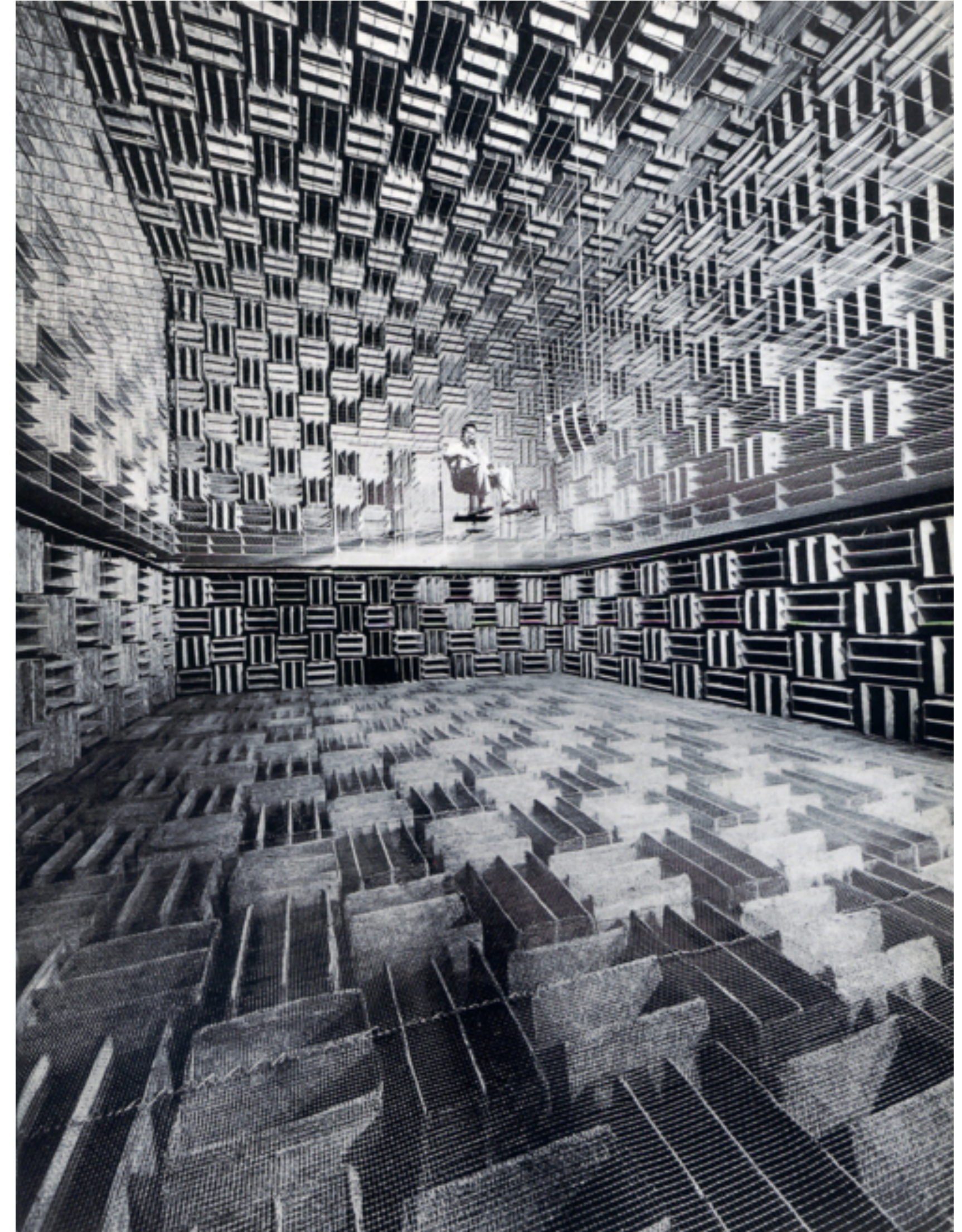
Reverberation

- What you hear in a room is a sound and all of its echoes (and their subsequent echoes) as they reflect off surfaces



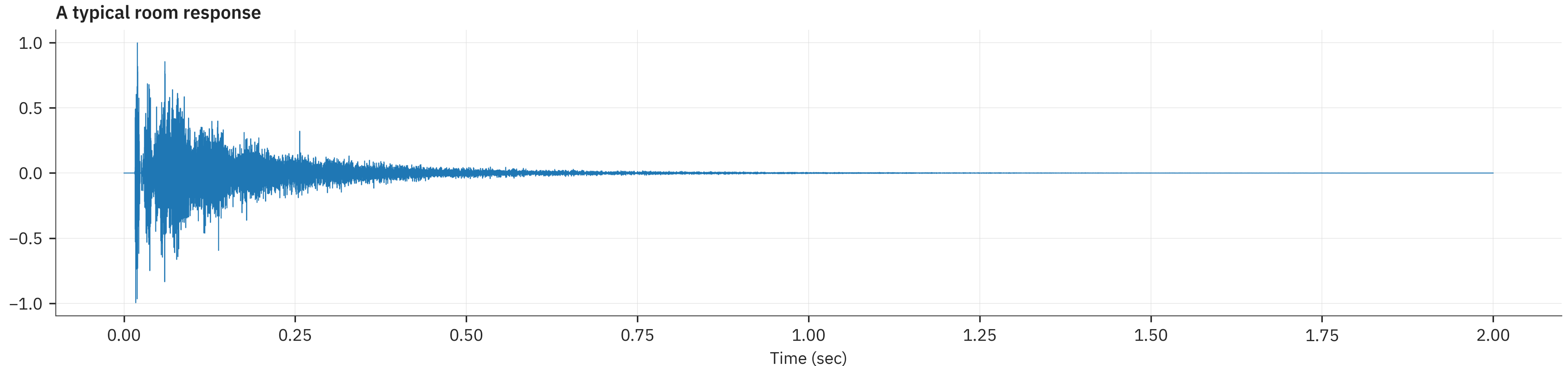
An exception

- Anechoic chambers
 - No noticeable reverberation due to special sound absorbing surfaces
- Very useful for acoustical testing and evaluation



Rooms as filters

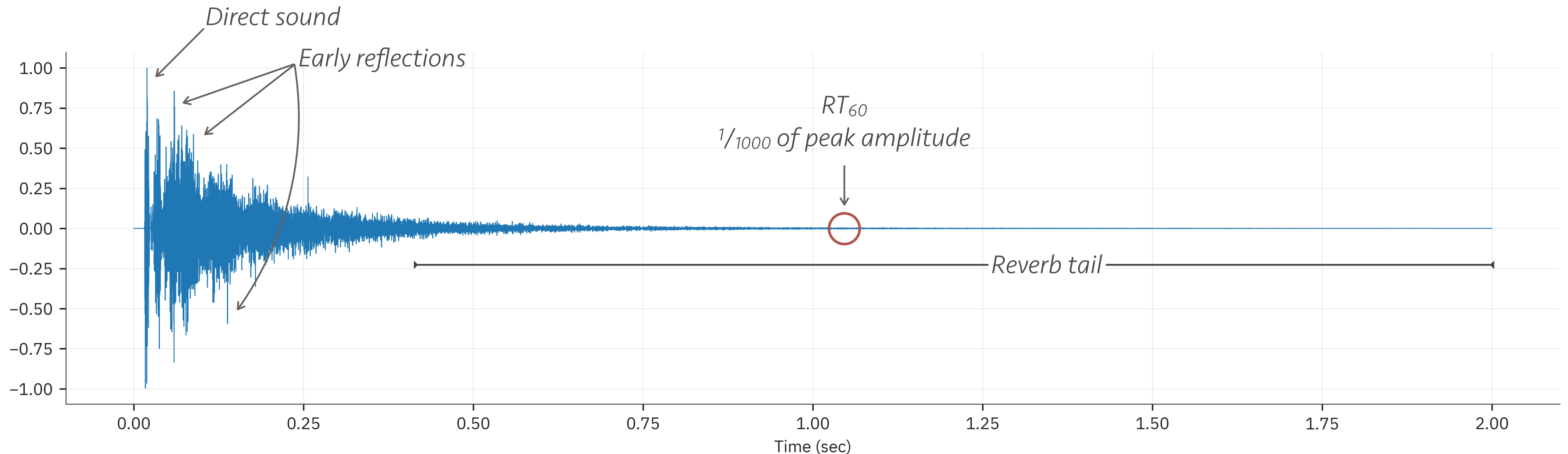
- Each sound reflection will cause an echo
 - We can create echoes with a convolution



- The set of these echoes is known as the *Room Impulse Response (RIR)*

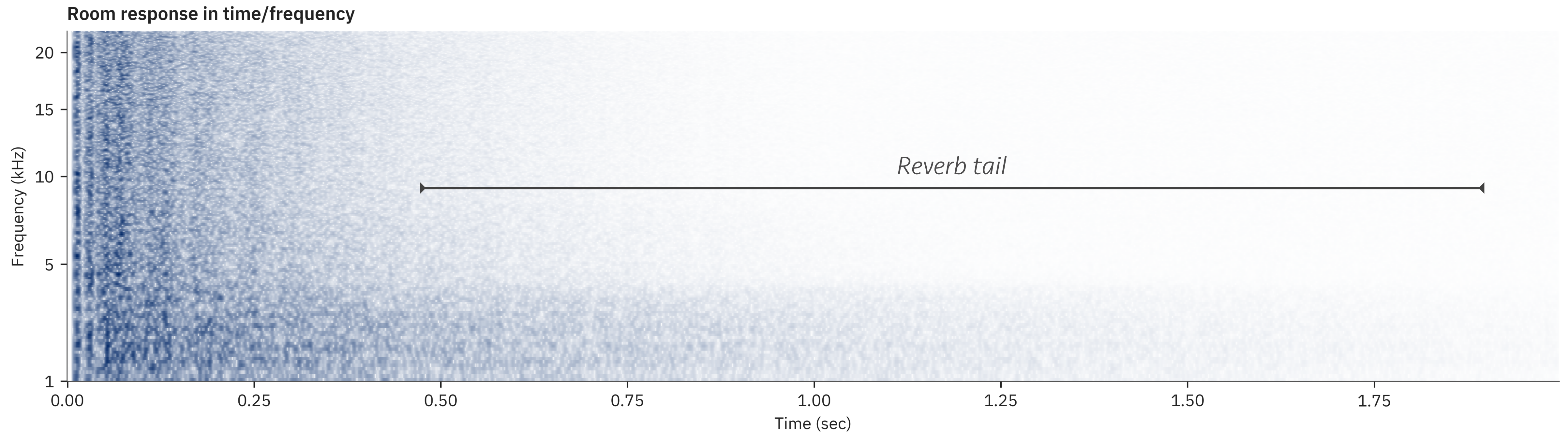
Anatomy of reverb

- Three main parts
 - Direct sound ; early reflections ; tail
- Main characteristic value: The RT_{60}



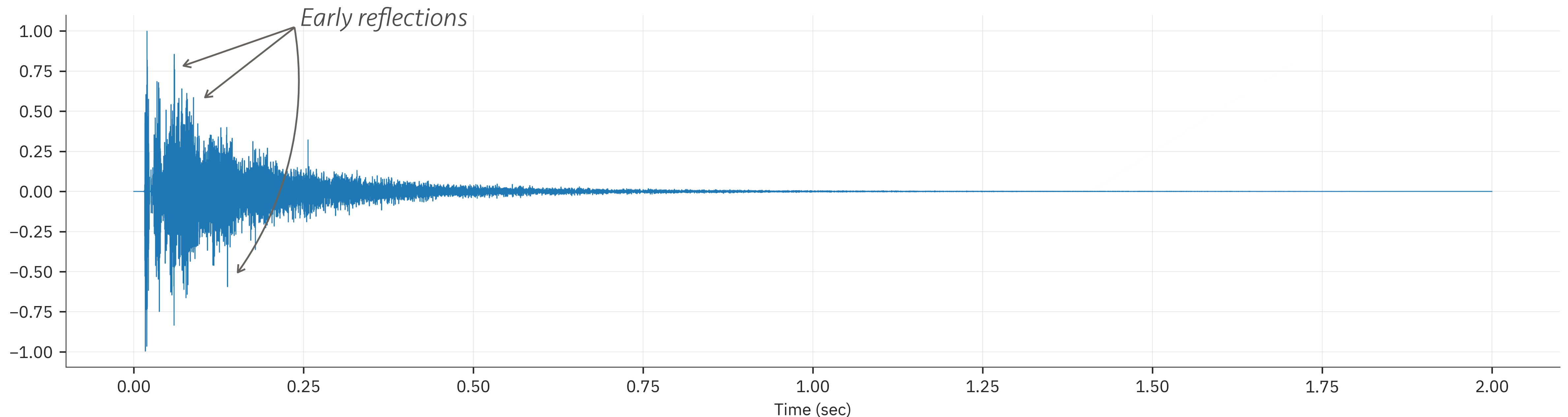
Anatomy of reverb

- Spectral view
 - Early reflections are generally broadband (why?)
 - Later echoes tend to muffle sound (why?)



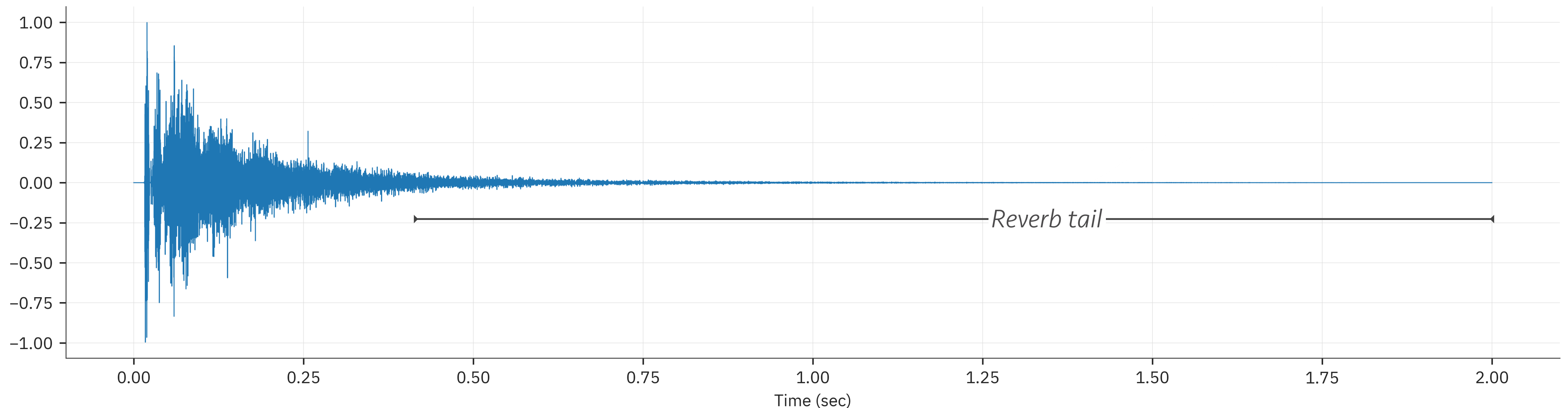
Early reflections

- First ~80-100ms in reverb response
 - Reflect structure of room
 - Key part to get right when emulating rooms



The tail (fused reflections)

- Almost looks like white noise
 - Echoes of echoes of echoes ... (1,000+ echoes/sec)
- Smoothly decaying energy



Some descriptors

- The Energy Decay Curve (EDC)
 - Measure how fast the reverb decays
 - The integral of the squared response over time (from t onwards)

$$\text{EDC}(t) = \int_t^{\infty} |h(\tau)|^2 d\tau$$

$$RT_{60} = t, \text{ where } \text{EDC}(t) = \text{EDC}(0) - 60\text{db}$$

- The Energy Decay Relief (EDR) curve
 - Energy at time t for frequency ω

$$\text{EDR}(\omega, t) = \int_t^{\infty} |H(\omega, \tau)|^2 d\tau$$

Some more

- Clarity index

- Amount of early reflections vs. reverb tail energies

$$C = 10 \log_{10} \frac{\int_0^T |h(t)|^2 dt}{\int_T^\infty |h(t)|^2 dt}$$

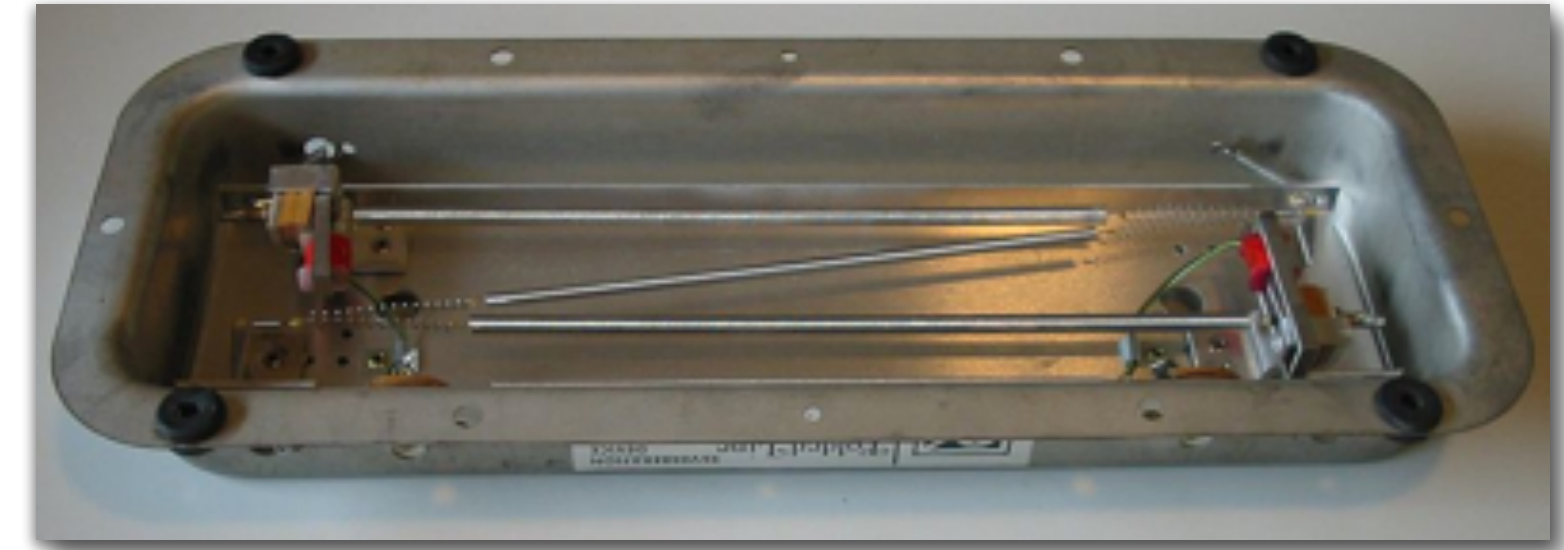
- Resonances

- For frequencies with wavelengths comparable to the size of the room we will get standing waves that create resonances
 - Roughly below $2000\sqrt{RT_{60}V}$ where V is the room volume (in m^3)

Simulating reverb

Clean signal

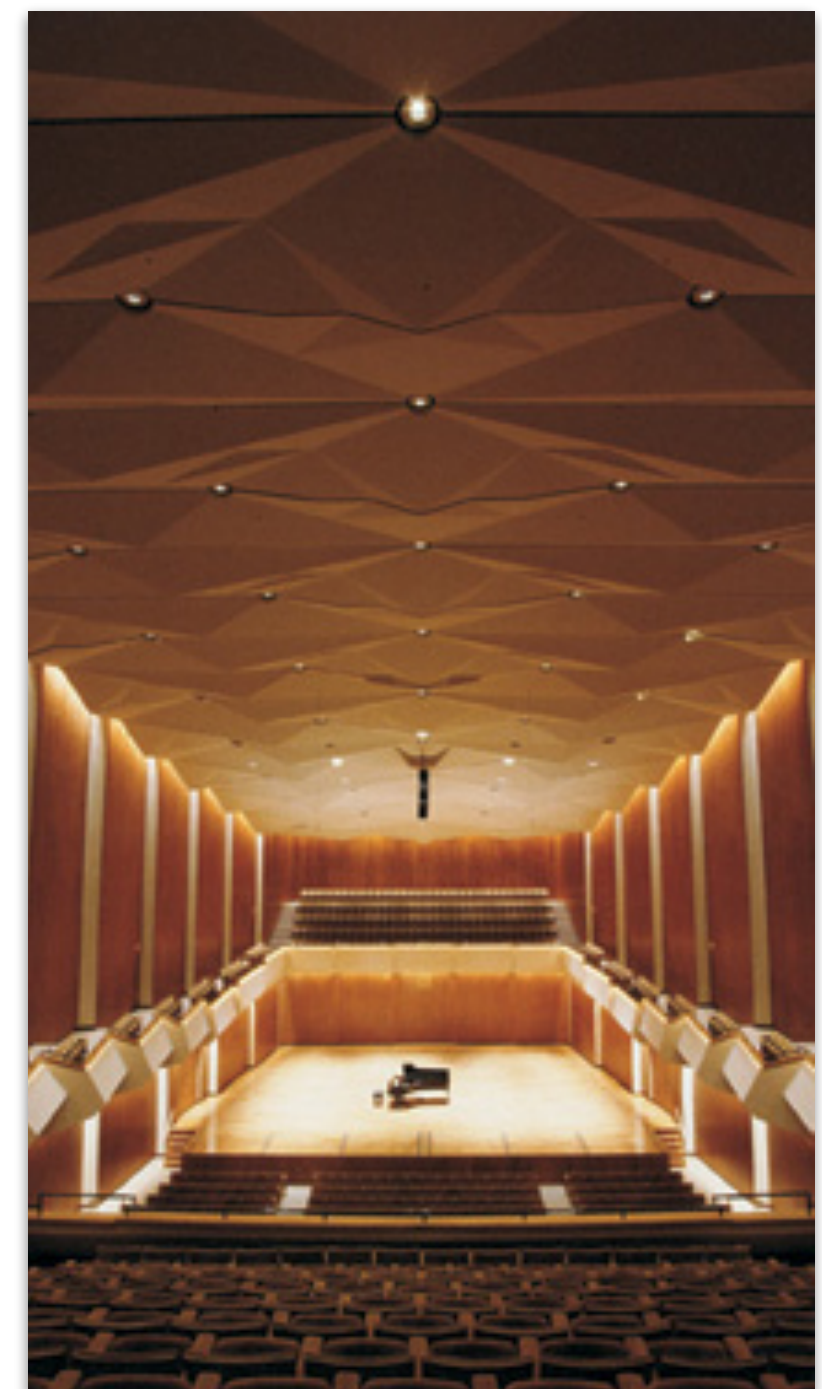
- Analog reverb simulators
 - The spring reverb
 - Use a vibrating spring to add reverb
 - The plate reverb
 - Use a resonating plate instead
 - Adjustable rooms
 - Very expensive!!



With spring reverb



Real-life example

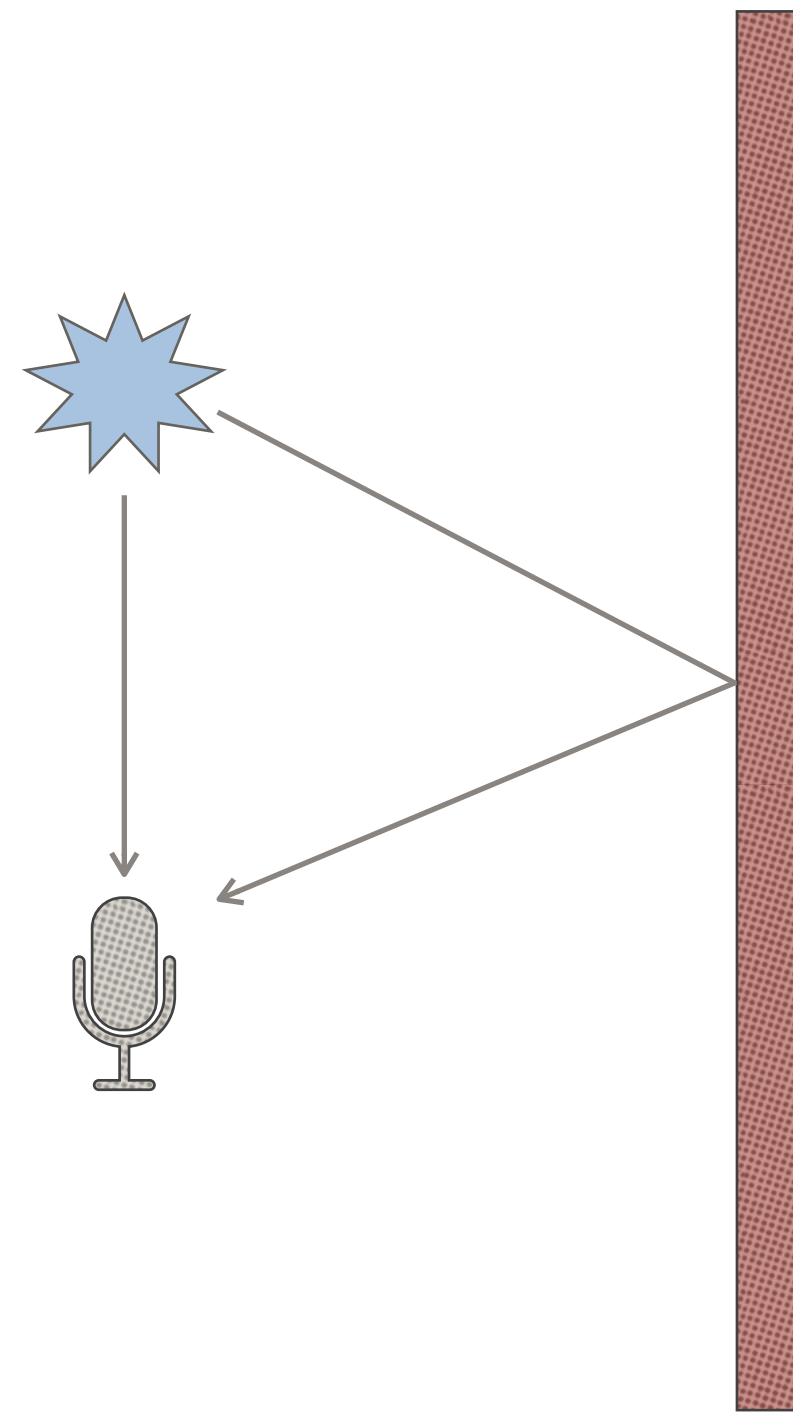


Simulating reverb digitally

- We can emulate a room by matching the early reflections and the reverb tail
- Two main approaches
 - 1) Fake it with filters
 - Fast and easy
 - 2) Compute impulse response from room geometry
 - Computationally heavy, needs some more thinking

Modeling a very simple room

- How will this room's impulse response look like?



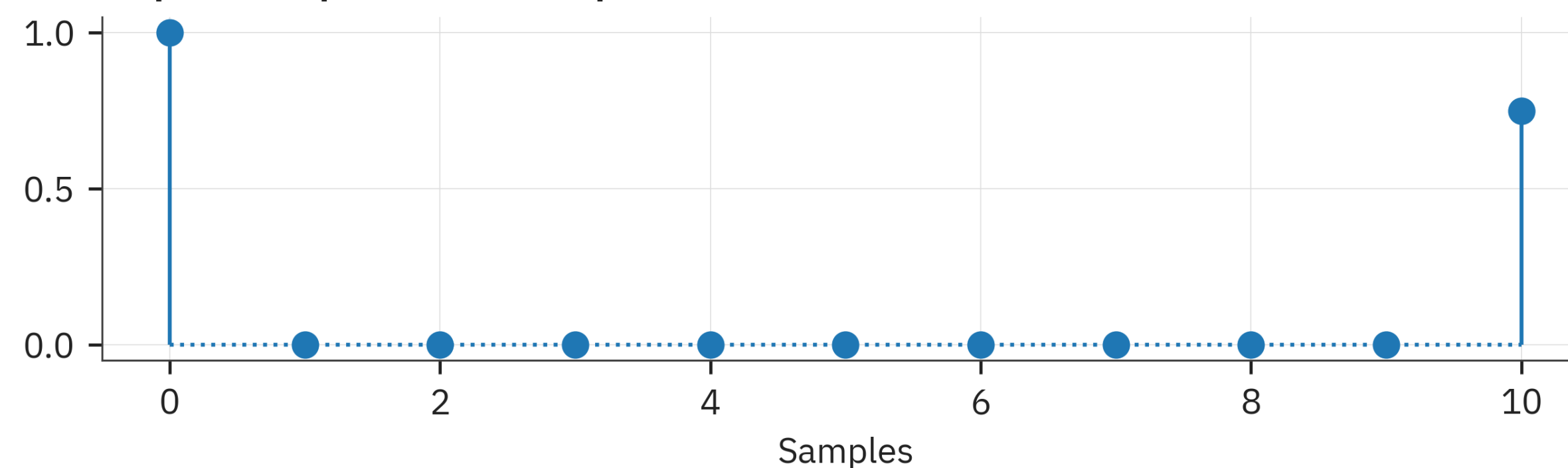
Modeling a sole echo

- FIR filter will be:

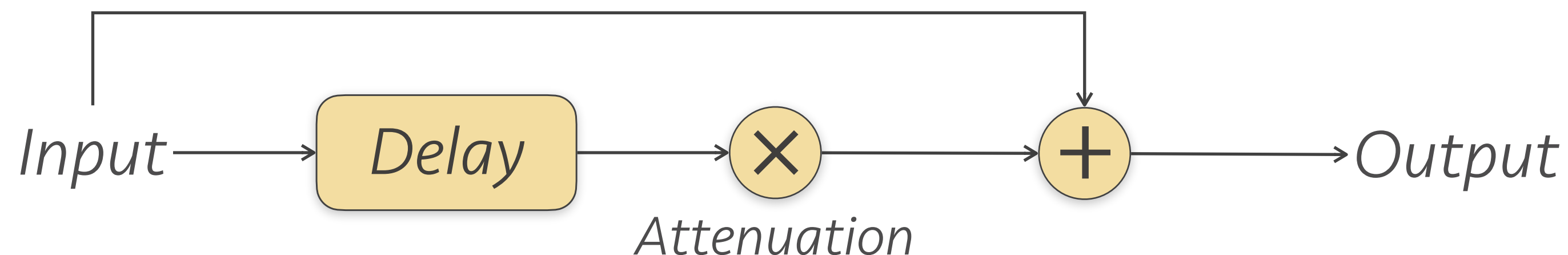
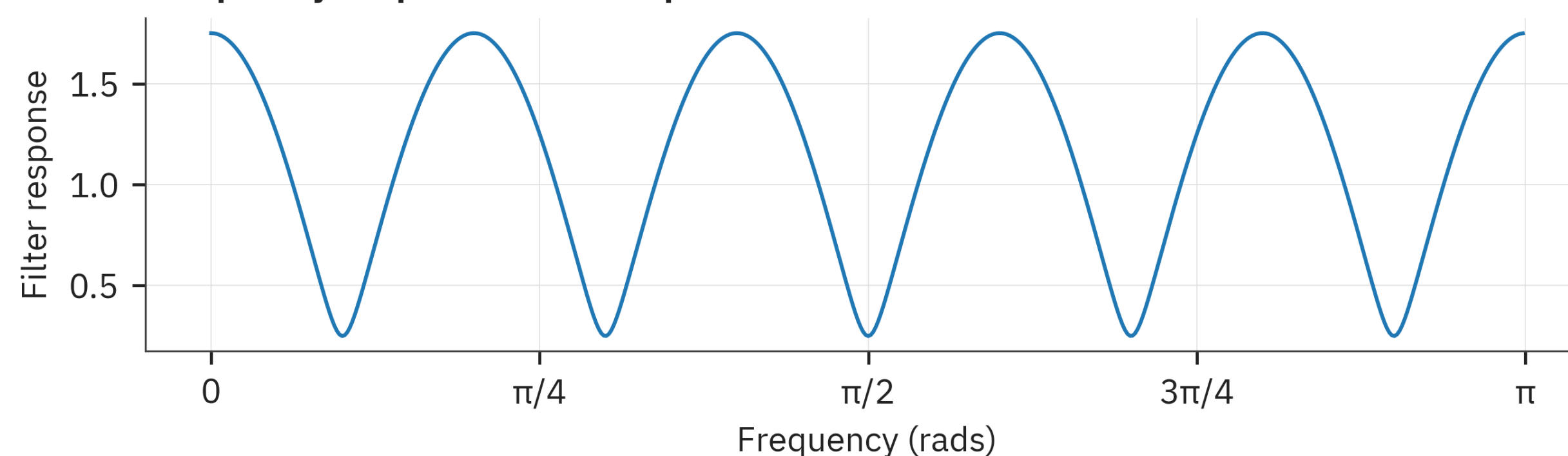
$$y[t] = x[t] + g x[t - N]$$

$$y = [1, 0, \dots, 0, g] * x$$

Impulse response for a simple echo

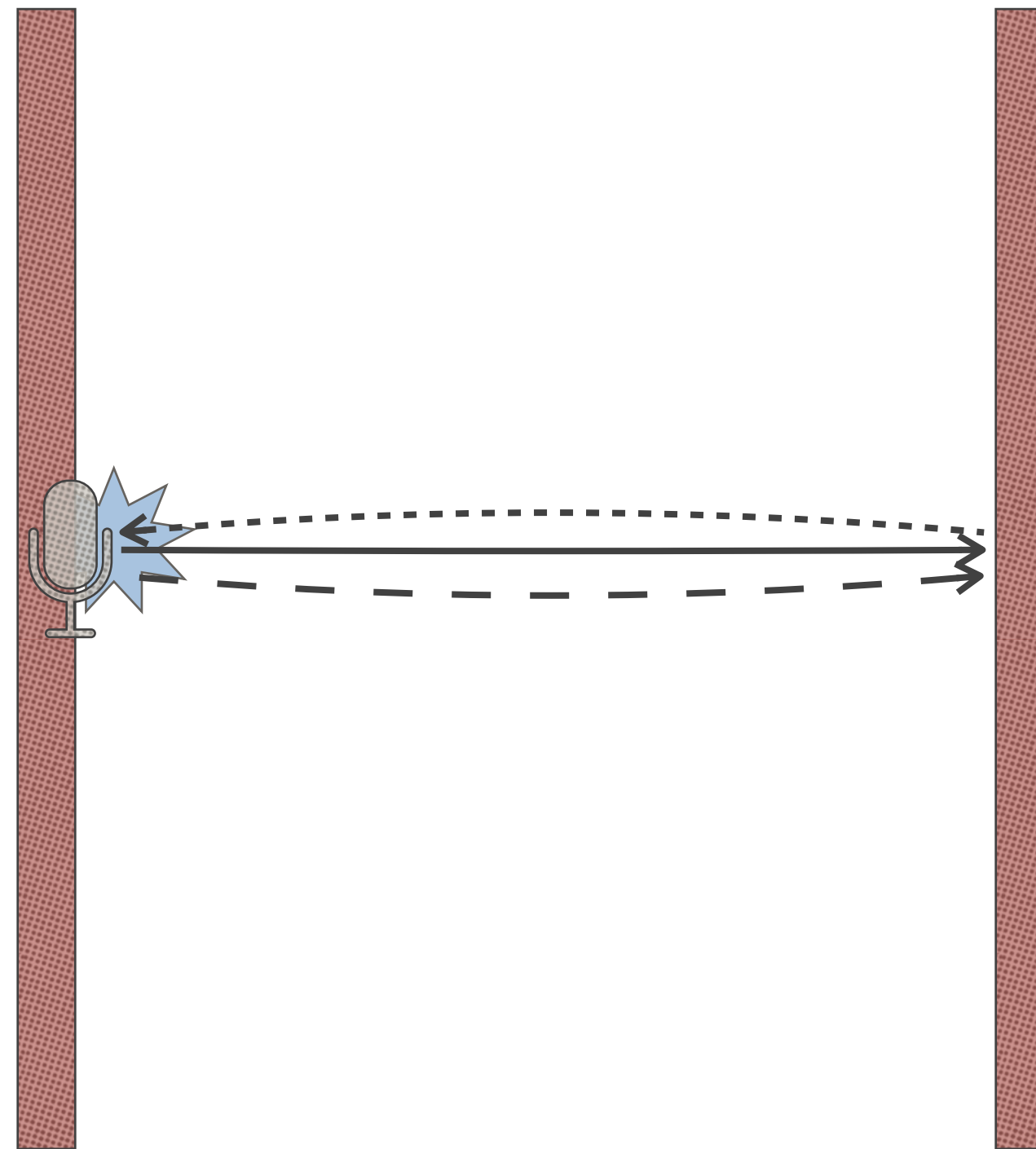


Frequency response for a simple echo



One step up

- How will this room impulse look like?

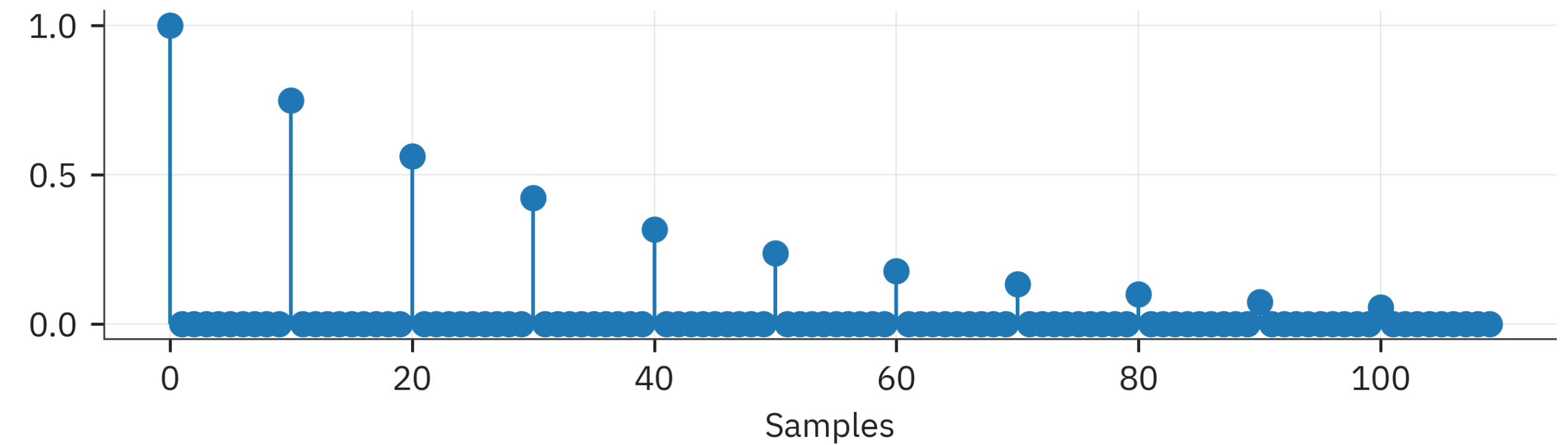


Comb filters

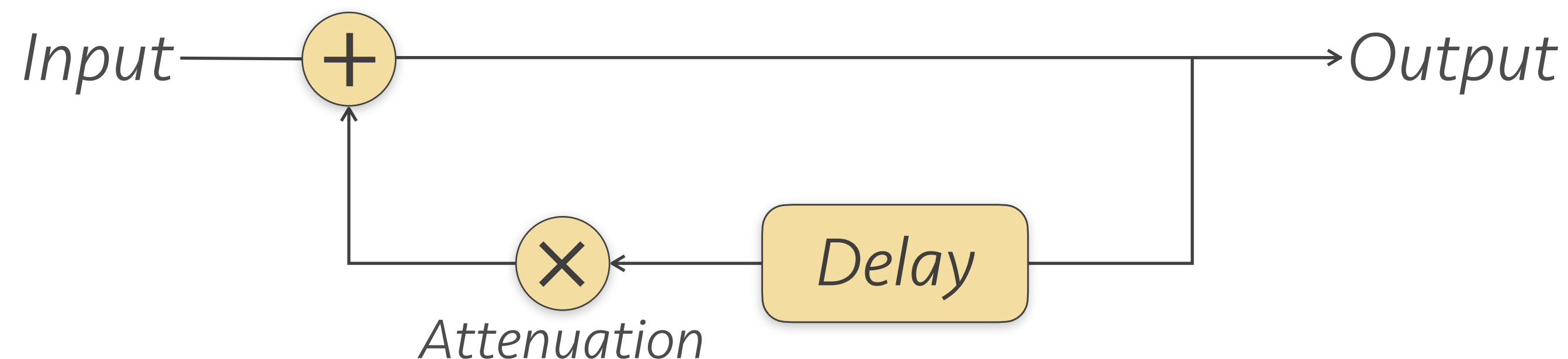
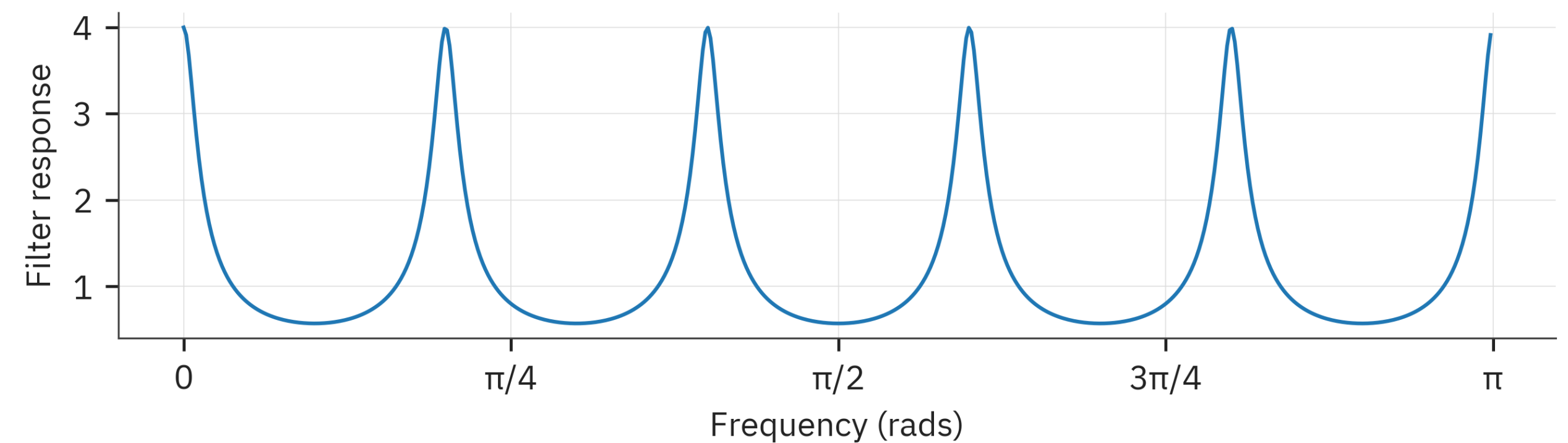
- We can model an infinite echo with:

$$y[t] = x[t] + g y[t - N]$$

Impulse response for a repeating echo



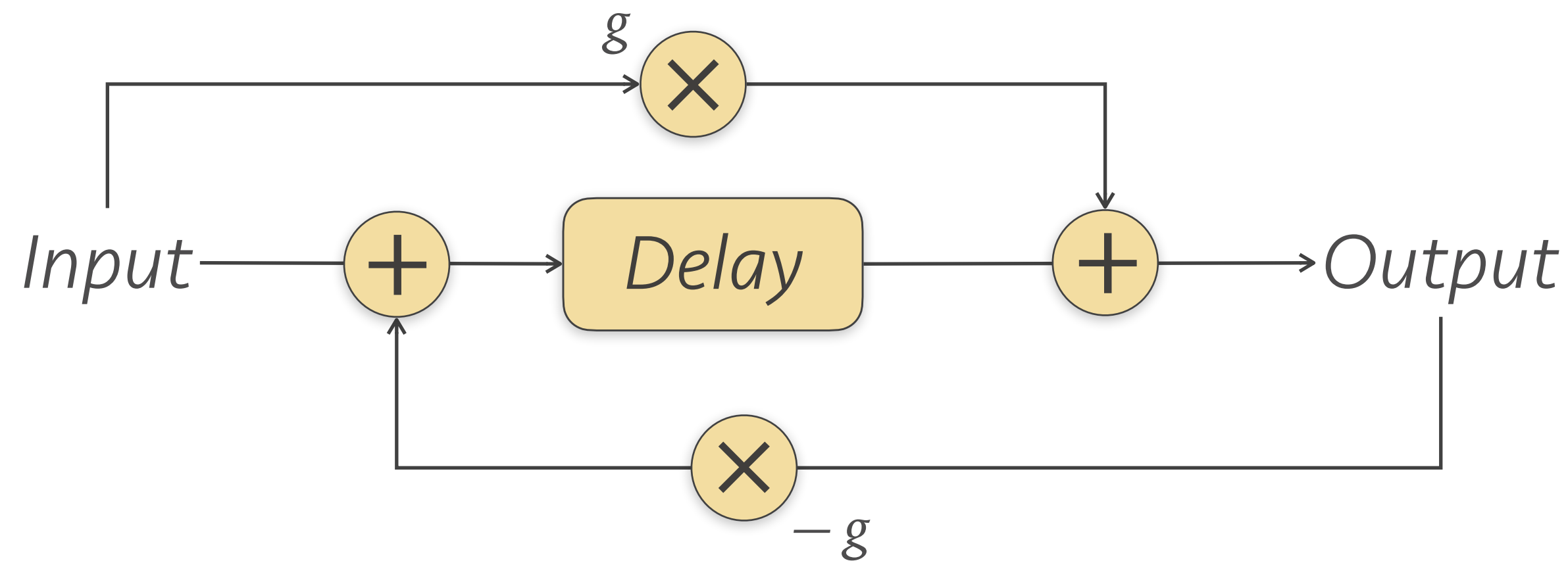
Frequency response for a repeating echo



Allpass filters

- Combines the previous two to avoid spectral coloring

$$y[t] = g x[t] + x[t - N] - g y[t - N]$$

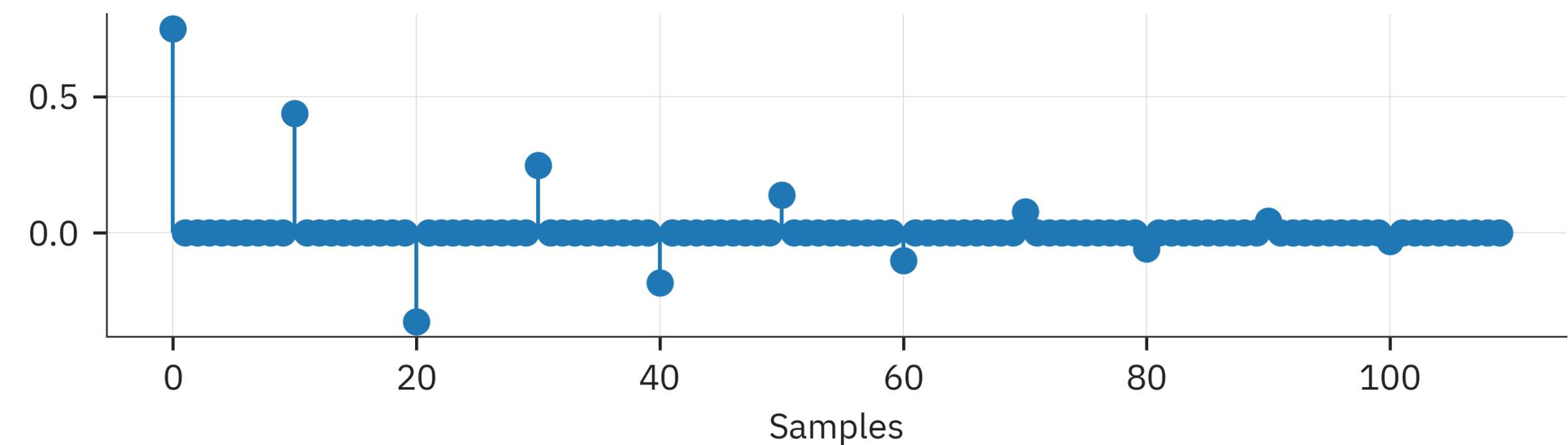


*Spectral
coloring
effect*

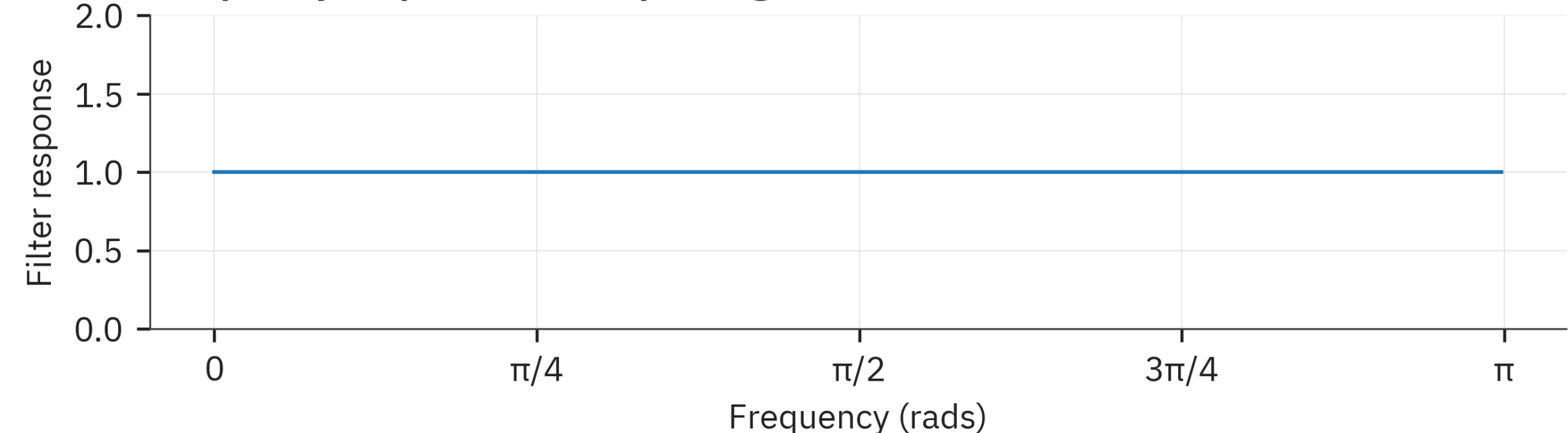
Comb filter

Allpass filter

Impulse response for a repeating echo

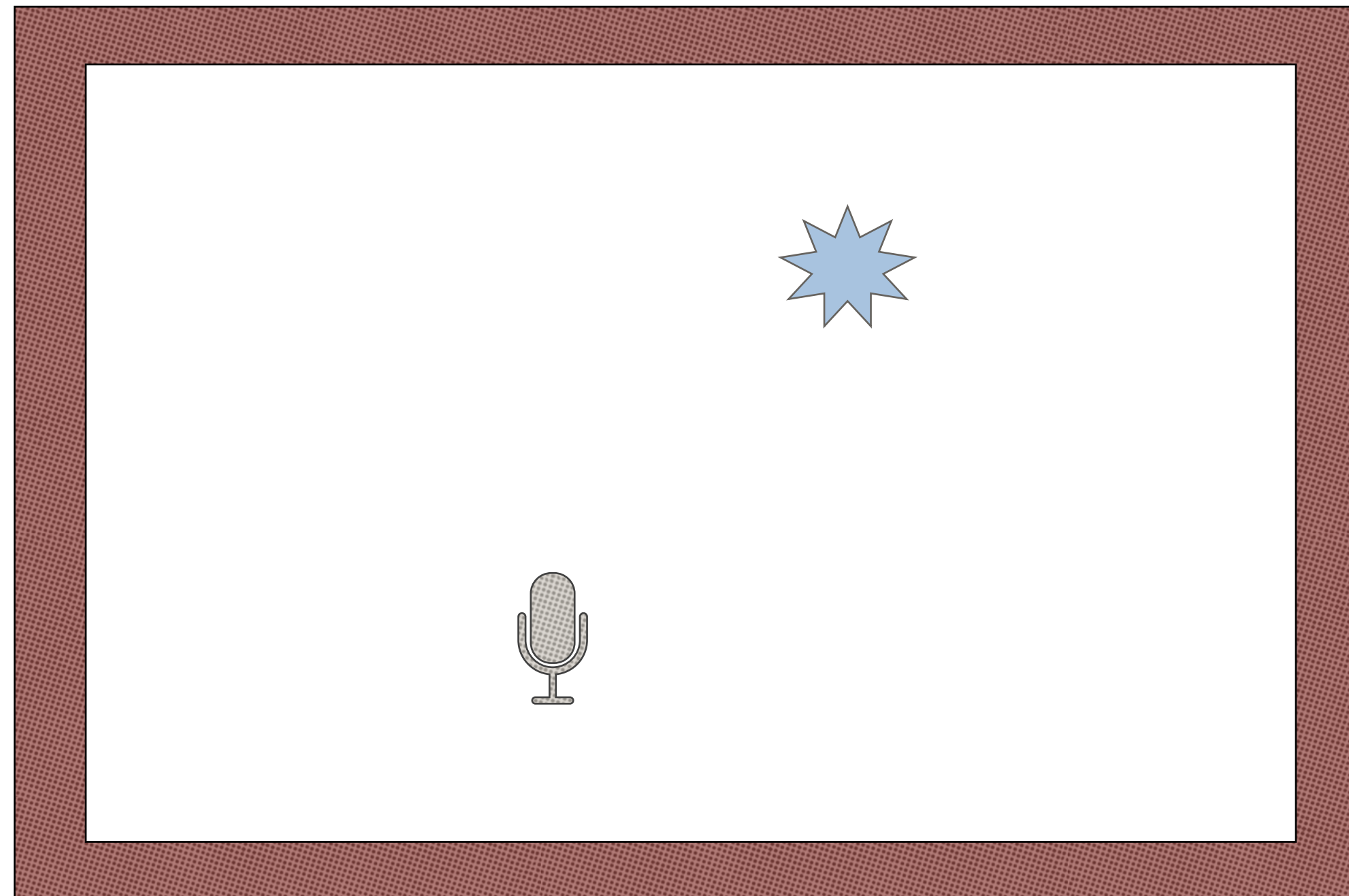


Frequency response for a repeating echo



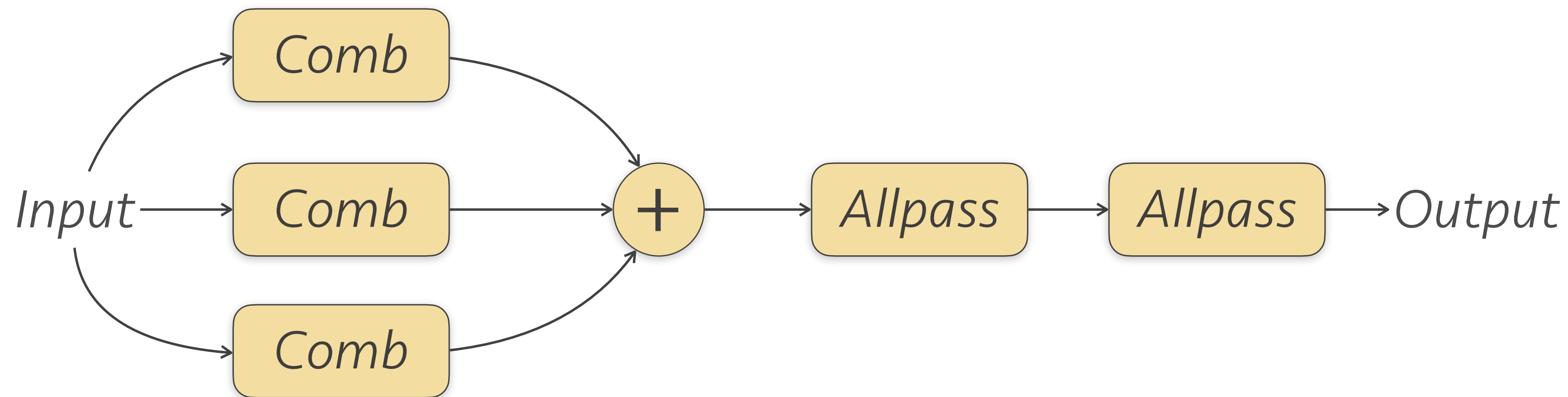
How about a room with many walls?

- We can have multiple walls in a simple room
 - Many different orders of reflections



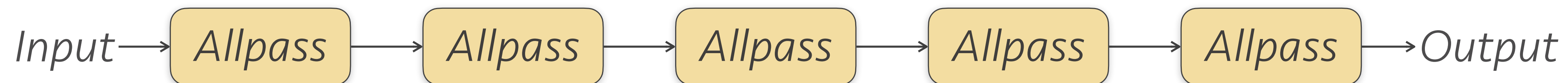
The Schroeder model

- Combine multiple comb/allpass filters
 - Parallel comb filters model ERs and add "coloring"
 - Allpass filters in series perform diffusion
- All filters should have unrelated delays!



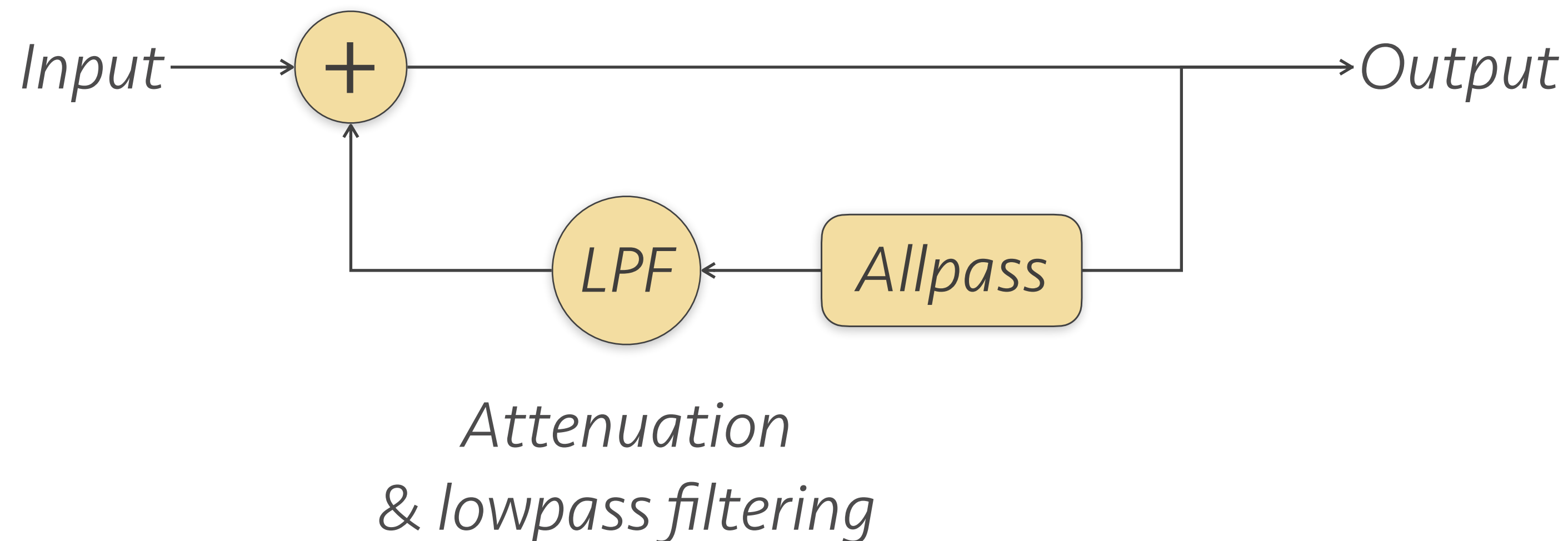
The other Schroeder model

- Combining multiple allpass filters in series
 - No ER model from comb
 - Smoother sounding, very diffuse
 - Allpass filters in series are also called diffusers



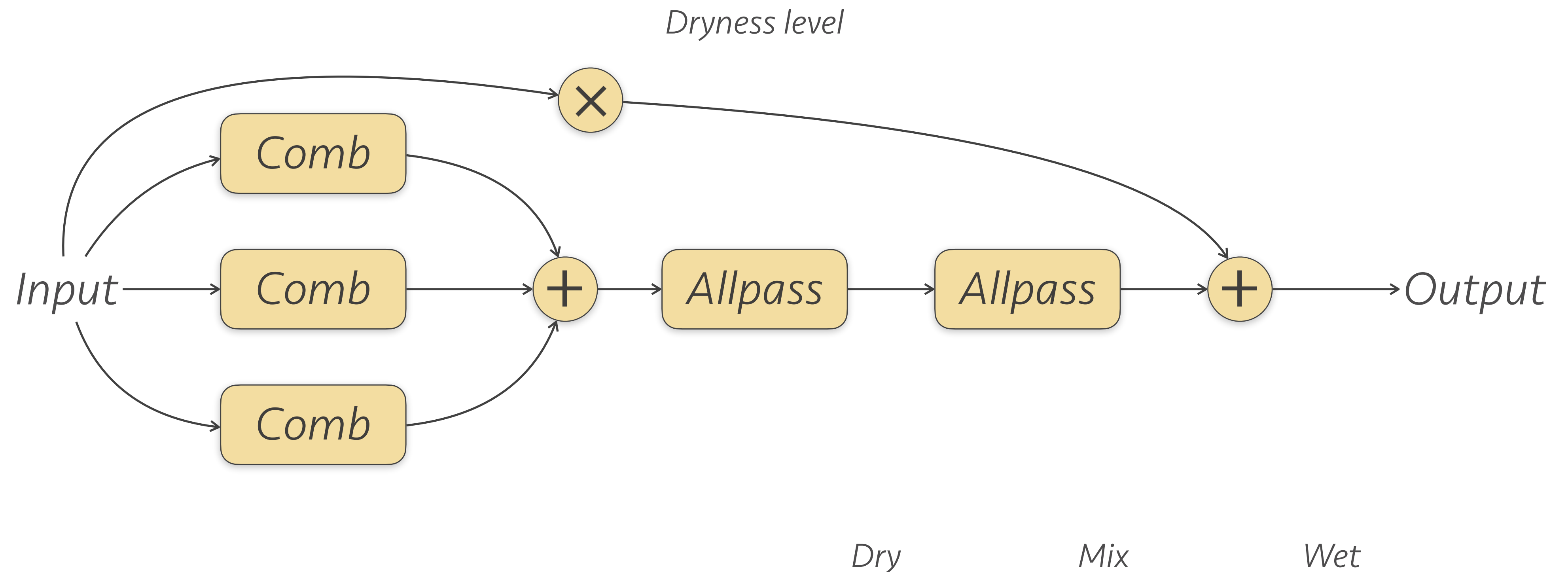
Elaborations of the Schroeder model

- We can also account for scattering
 - Add a lowpass filter in the comb structure
 - We can also do the same with the allpass structure



Wetness of reverb

- “*Wet*” means a lot of reverb
- “*Dry*” means no reverb



Shortcomings and issues

- Can sound too synthetic
 - Picking unrelated delays for the filters is key!
 - Schroeder's suggestion:

Unrelated delays

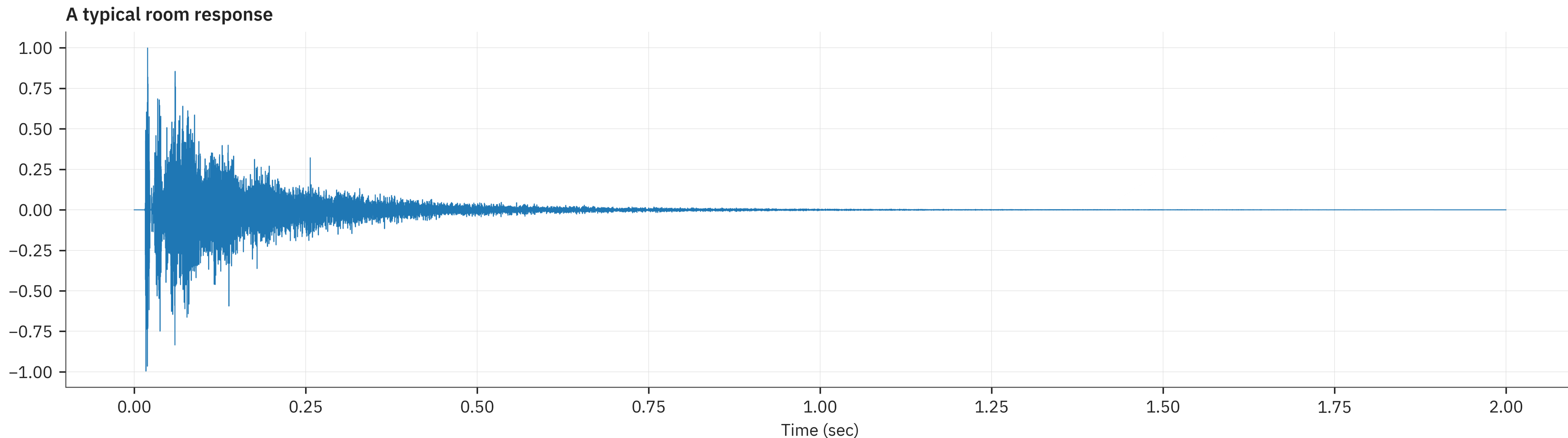
$$N_i = \frac{F_s}{10 \cdot 3^i}$$

Related delays

- Not really a room model
 - Doesn't account for room geometry or acoustical effects

Reverberation via convolution

- Since the room is a filter, we can directly convolve with the room's impulse response
 - Advantage: Exact model
 - Disadvantage: What's the room's impulse response?



Approach 1: Cheating

- We can make up a room response
- Things to look for:
 - First 100ms model the early reflections
 - Need a few prominent echoes, no coloring, not too busy
 - Rest of response needs a very dense echo count
 - Rule of thumb: 1,000+ echoes per sec

Making up responses

- Early reflections
 - Can use sparse noise
 - E.g., Cauchy noise (divide two Gaussian noise sequences)
- Reverb tail
 - Exponentially decaying noise
 - Make sure it's white to avoid coloring
 - Or not, if you want coloring!

Using real room impulse responses

- Instead of making them up, use real responses
- Two ways to do this:
 - Go download one (e.g. [here](#))
 - Estimate the response of a room
 - Either analytically, or by measuring recordings in it

Measuring room impulse responses

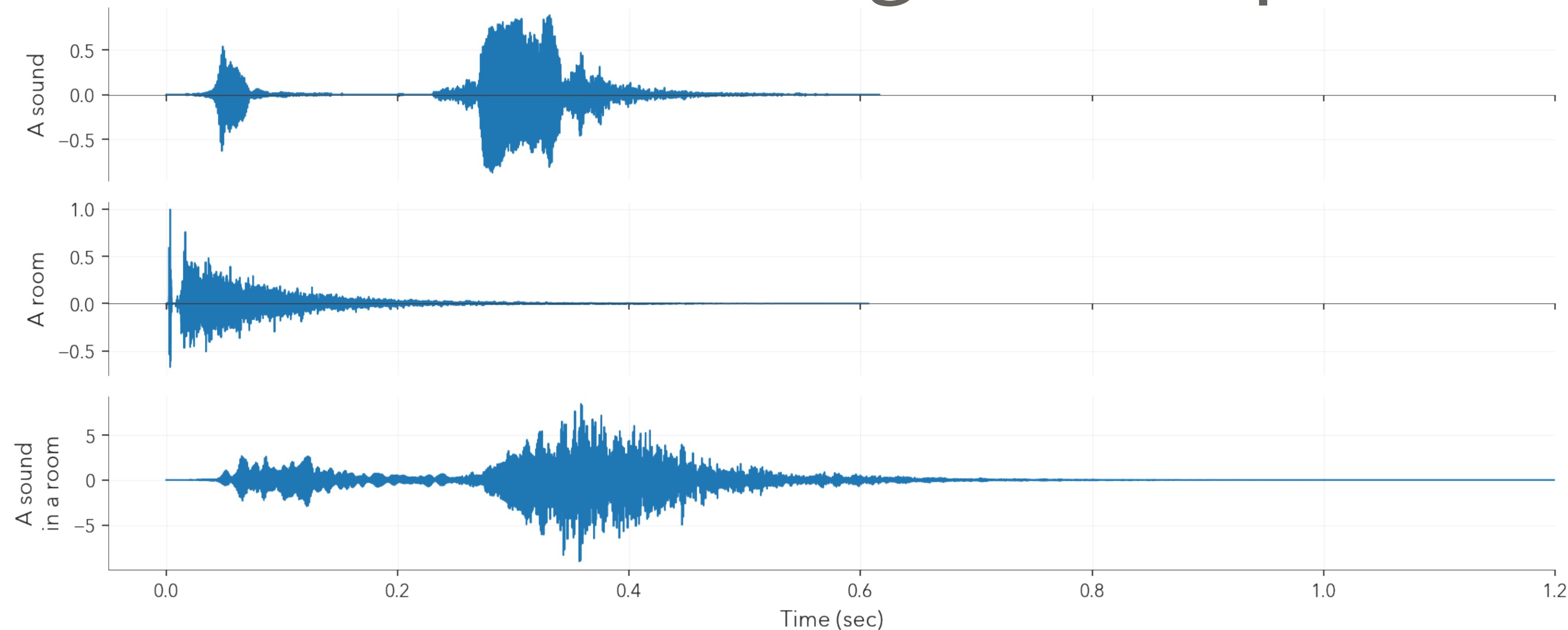
- The direct approach
 - Make an impulsive sound in the room (clap, balloon pop, gun)
 - Measure the resulting sound



<https://www.youtube.com/watch?v=ILUcOFwZvyY>

How does that work?

- Assume that the sound we make is $[1, 0, 0, 0, 0, 0, \dots]$
 - Output: $y[t] = h[t] * [1, 0, 0, 0, \dots]$
 - Simulate with: $z[t] = y[t] * x[t]$
- We can use the recording as an impulse



Some problems

- Extra noises will bias the measurement
 - What if someone drops a coin while recording?
 - You will get an extra metal-sounding echo!
- Hard to get an exact impulse sound
 - Excitation will most likely be “colored”
 - Will result in a biased RIR estimate

A more robust approach

- Deconvolution
 - Excite room with a more robust signal
 - Knowing the excitation, estimate the RIR
- Repeat the experiment multiple times
 - Average out the results for robustness
- Allows for more accurate RIR estimates

Deconvolution process

- Excitation is:

$$y[t] = h[t] * x[t]$$

$$Y[\omega] = H[\omega] X[\omega]$$

- Solve for RIR:

$$H[\omega] = \frac{Y[\omega]}{X[\omega]}$$

Convolution becomes element-wise
multiplication in the frequency domain
(Uppercase denotes DFT of variable)
t is time, *ω* is frequency

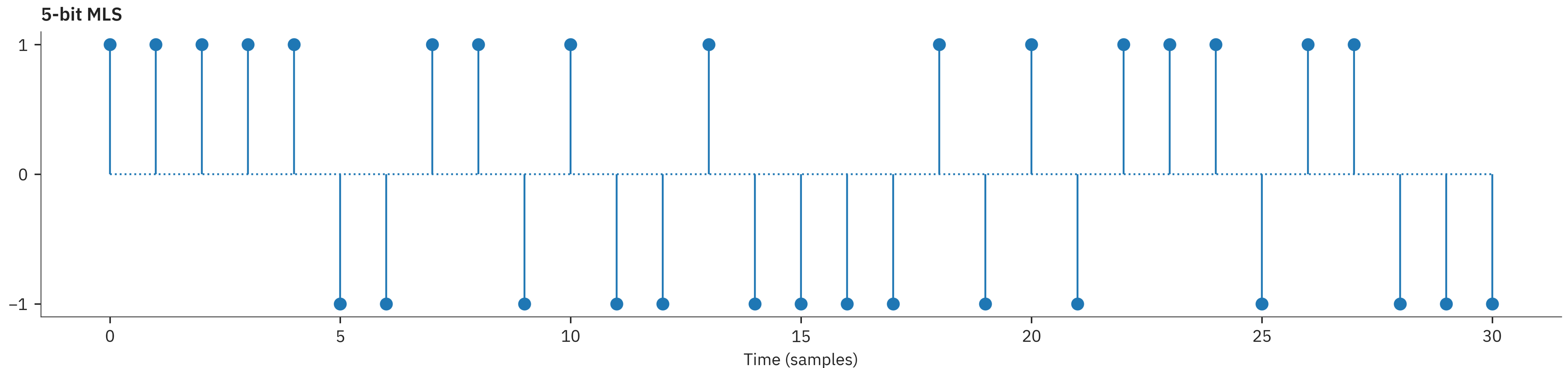
A better way

- *Maximum-Length Sequences (MLS)*
 - Sequences that *autocorrelate* well with themselves

$$X[\omega] X[\omega]^* \approx 1$$

* is complex conjugate
 $(a + \imath b)^* = a - \imath b$

- Binary sequences with length $2^N - 1$



Using MLS to measure RIRs

- Rewrite deconvolution as:

$$H[\omega] = \frac{Y[\omega]}{X[\omega]} = \frac{X^*[\omega] Y[\omega]}{X^*[\omega] X[\omega]} = X^*[\omega] Y[\omega]$$

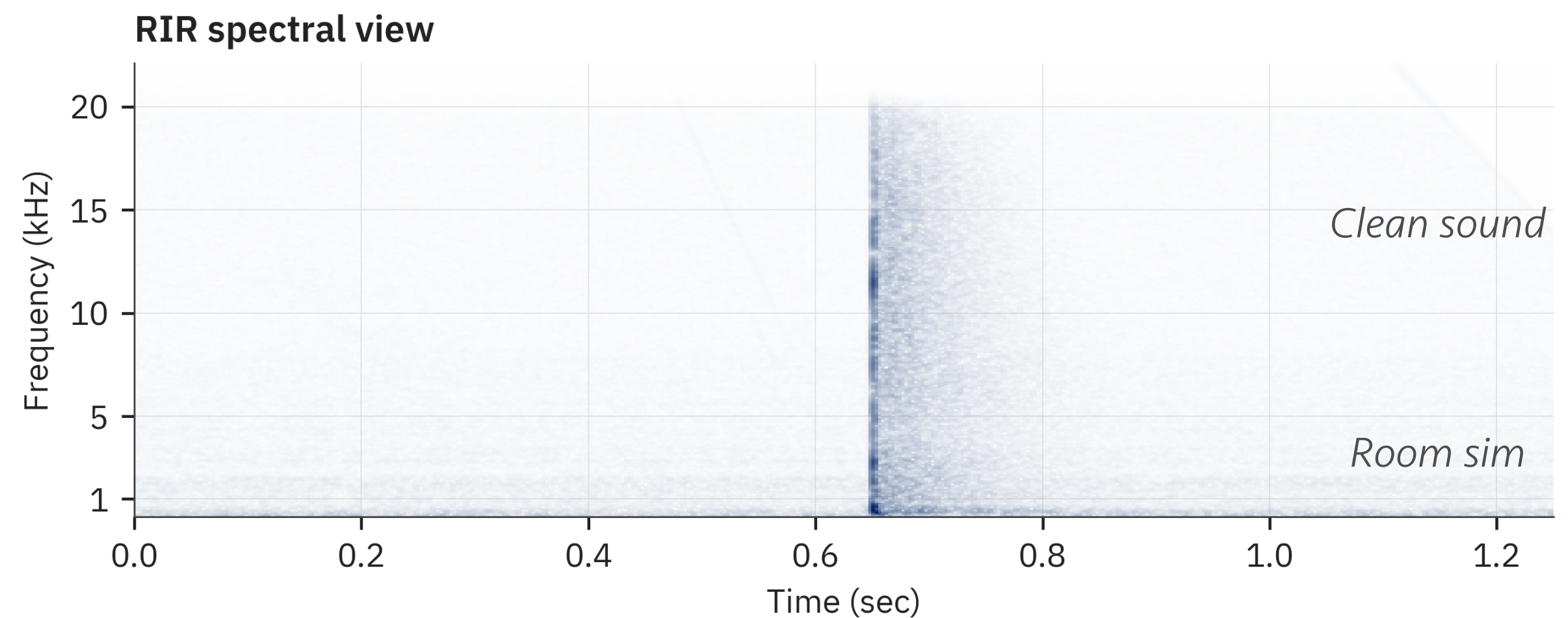
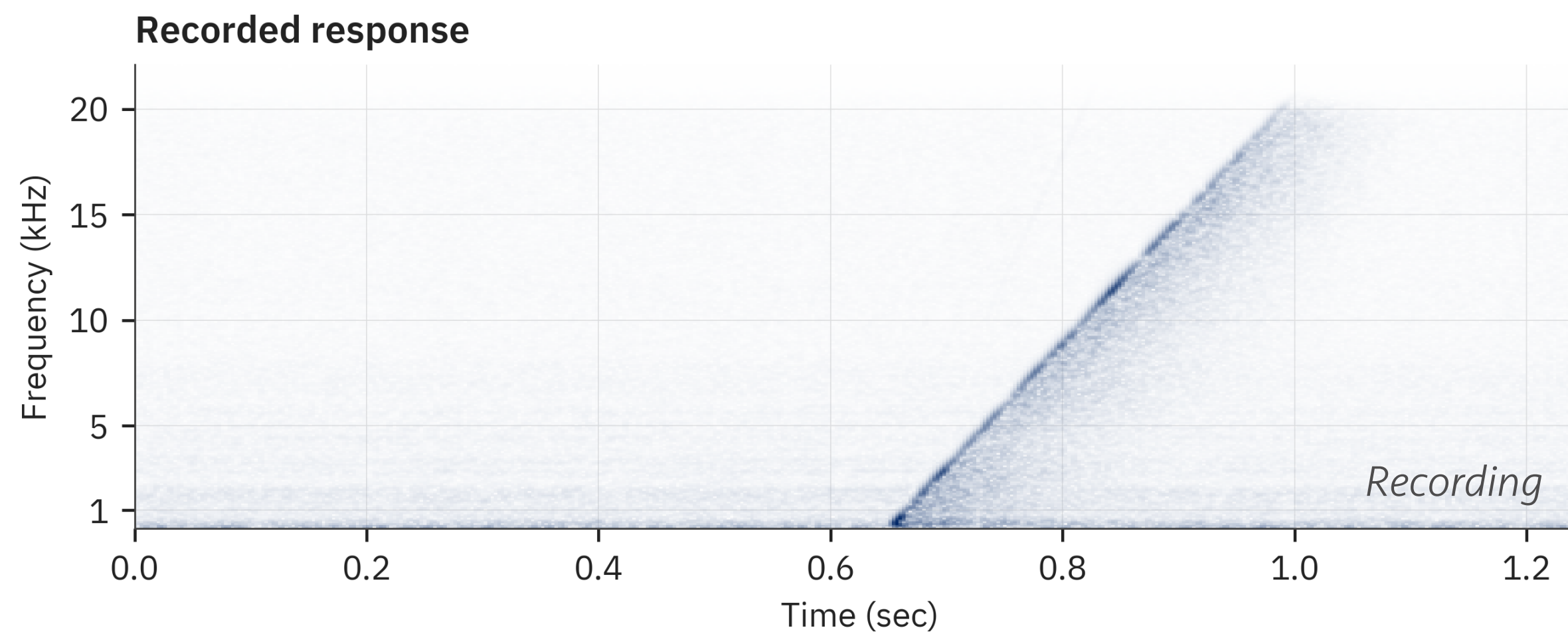
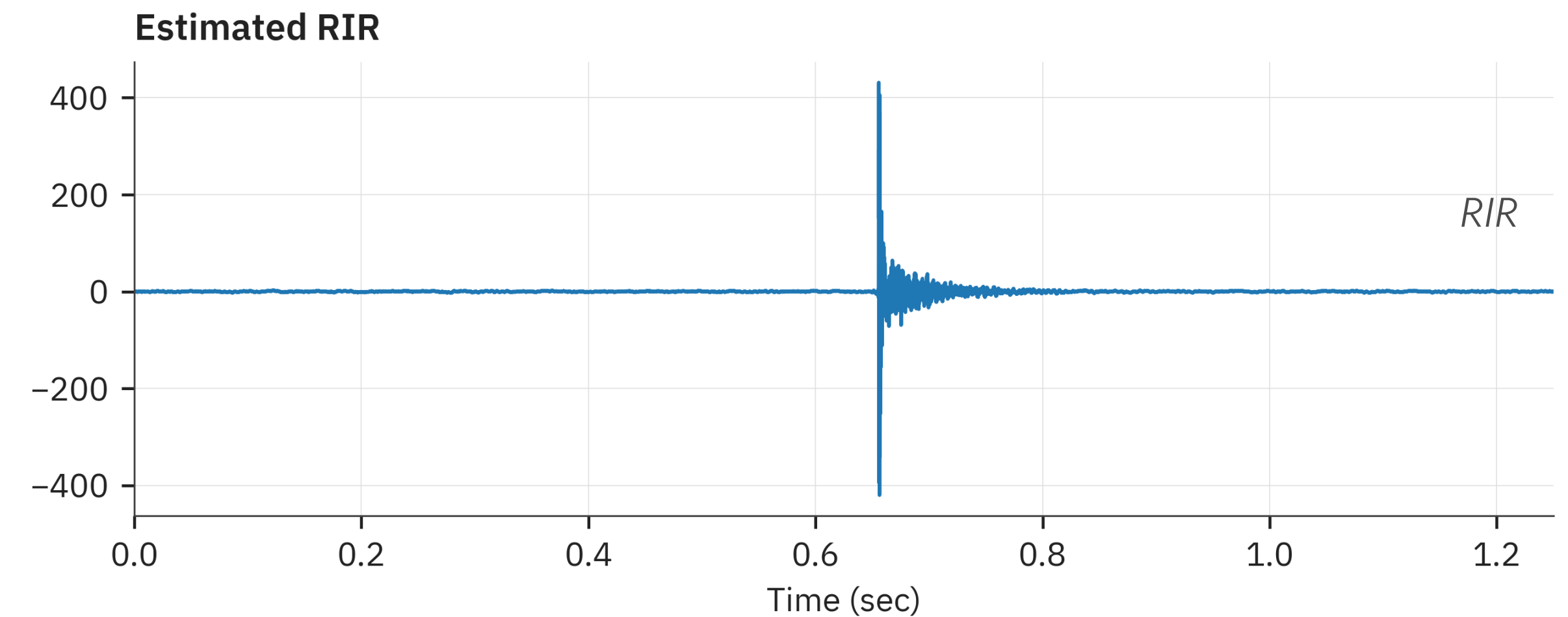
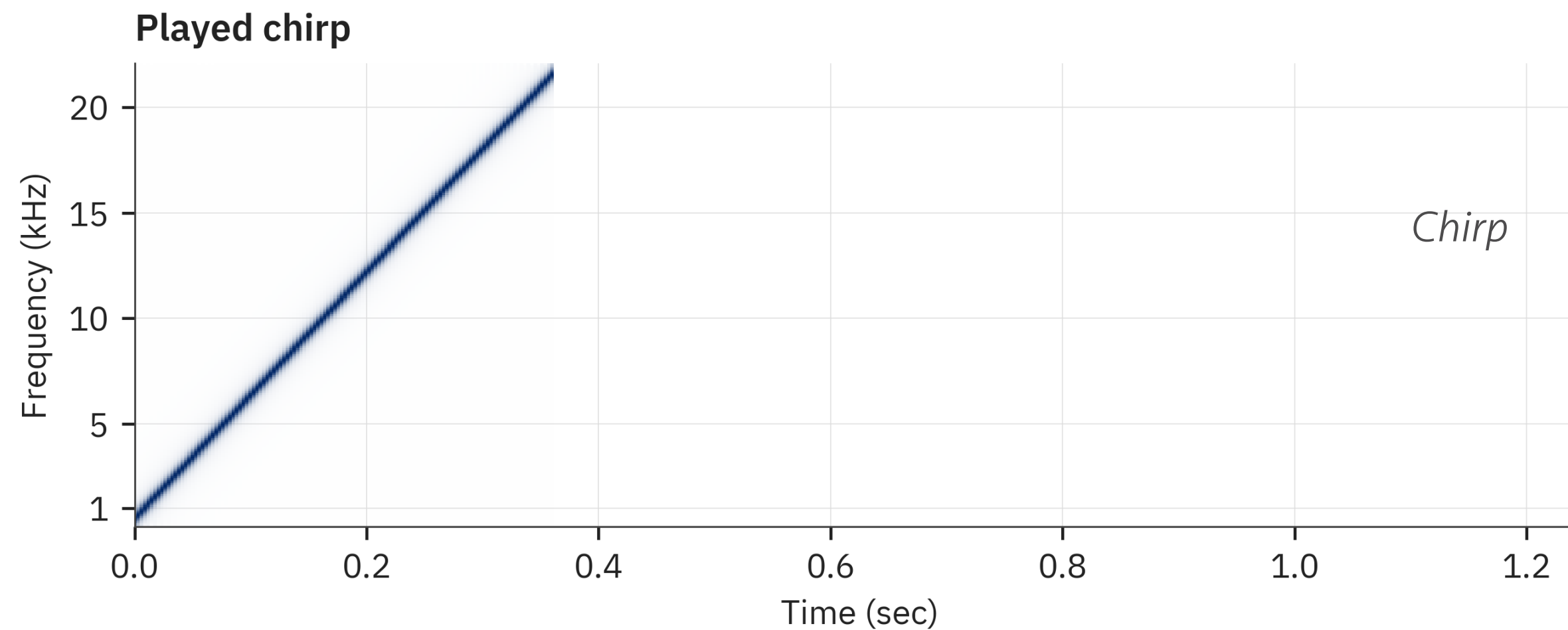
- Removes potentially problematic division
 - Fast and easy to compute
 - Also works with any other signals with that property

Alternative excitations

- The swept sine excitation (a chirp)
 - Pass through all frequencies individually
- The exponential swept sine
 - Has some better properties than swept wave
- White noise
 - Poor man's MLS!

Example case with swept sine

- Measuring my office's RIR



Tradeoffs and problems

- Longer excitations are better
 - More accurate spectra (more frequencies)
- Noise robustness via repetition
 - Average result out of many trials
- Be mindful of loudspeaker and mic responses!
 - You measure these as well

Physical models of rooms

- We can also make a model of a room
 - i.e., estimate the room response analytically
- Multiple approaches
 - Finite Element (FE) models
 - Image-source method
 - Path/Beam tracking

Tracing all paths

- Explicitly compute the delays from all echoes
 - Use them to construct the RIR
- How to we find the two delays here?

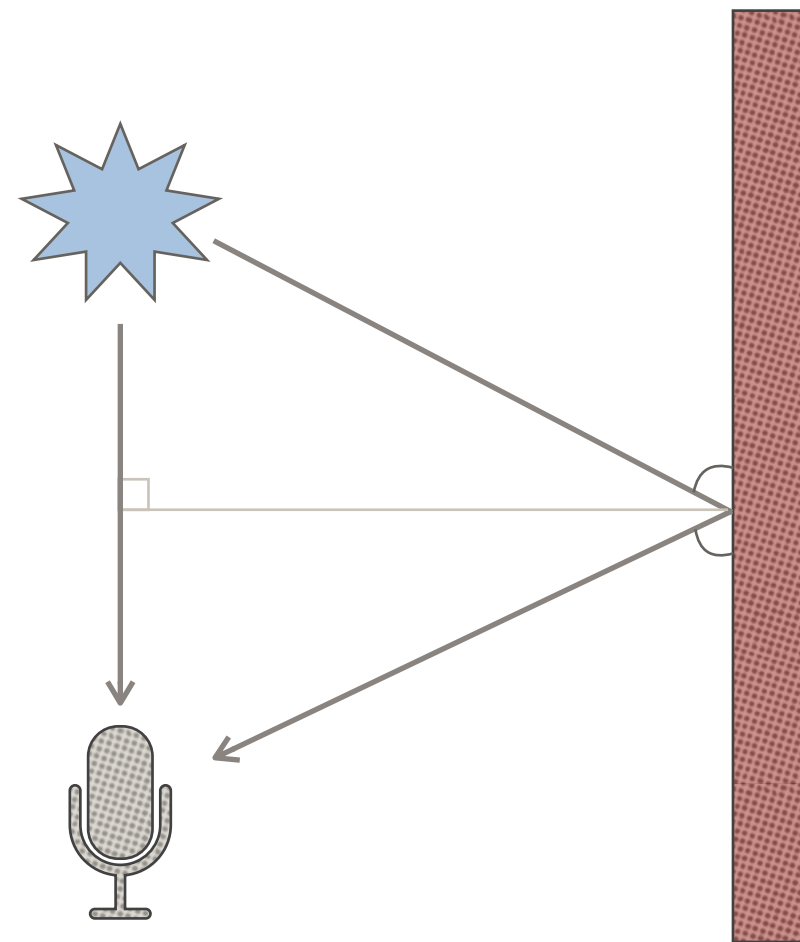
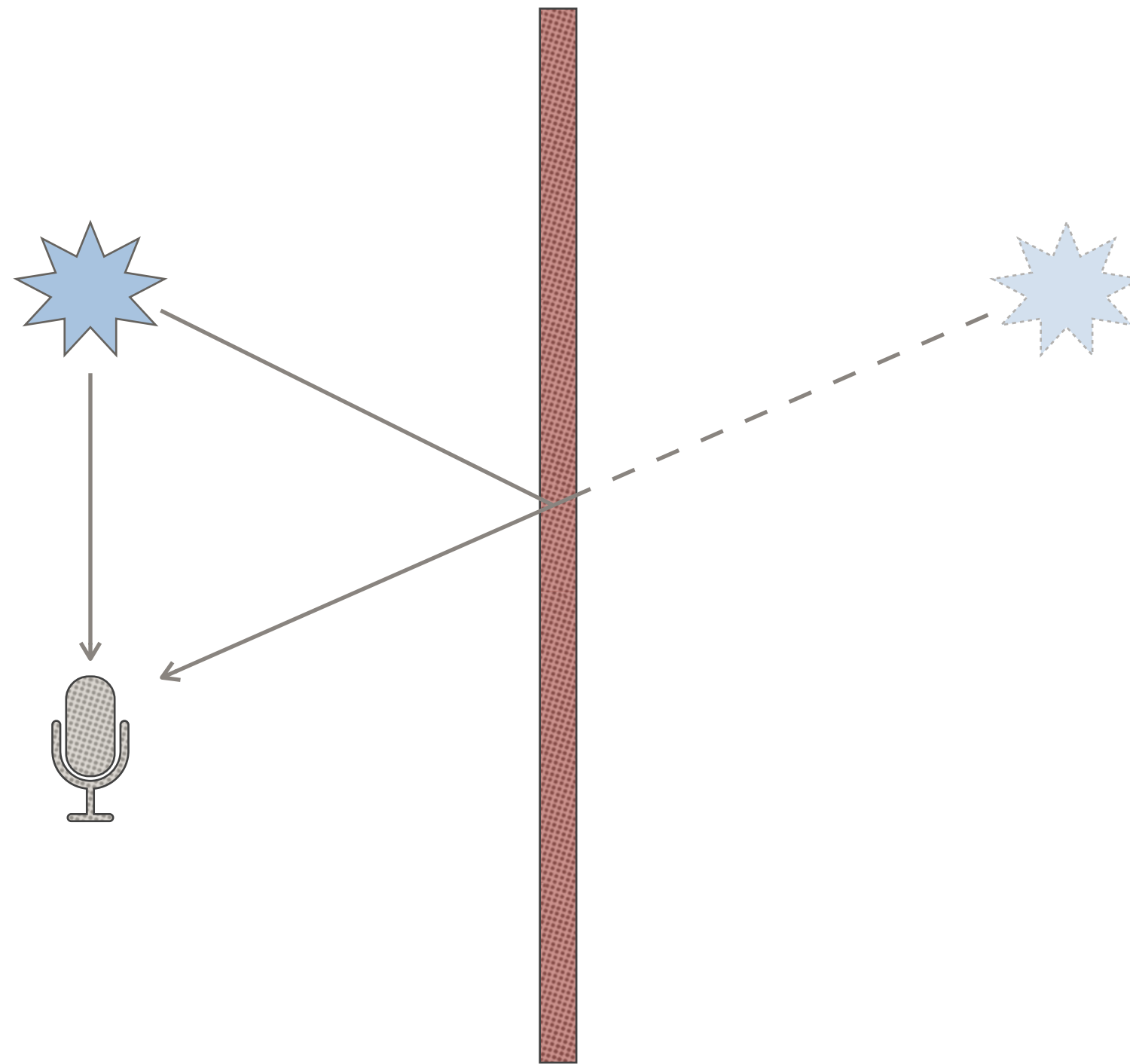
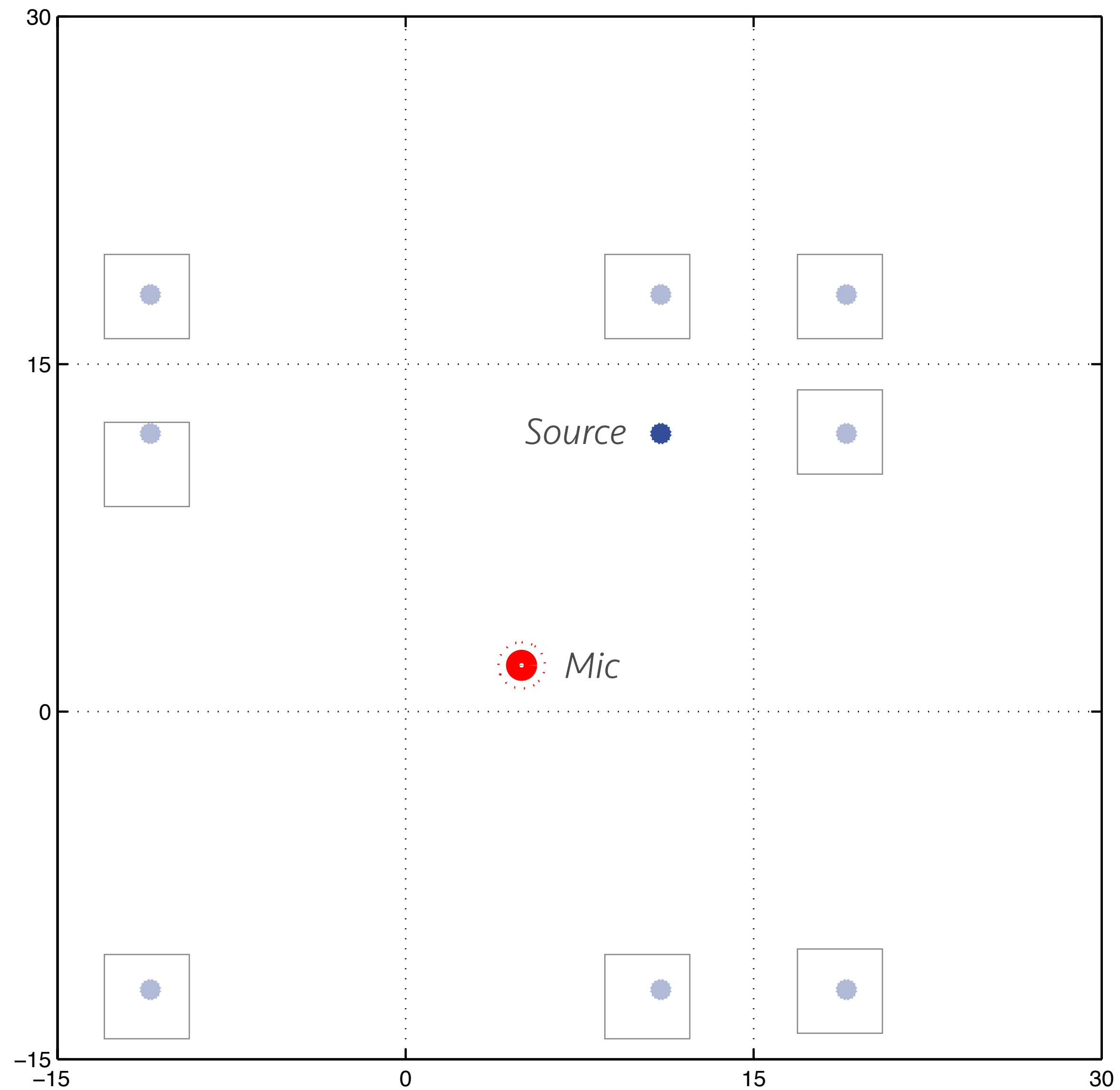


Image-source model

- Project source to a *virtual location* across the wall
 - Calculate that distance instead



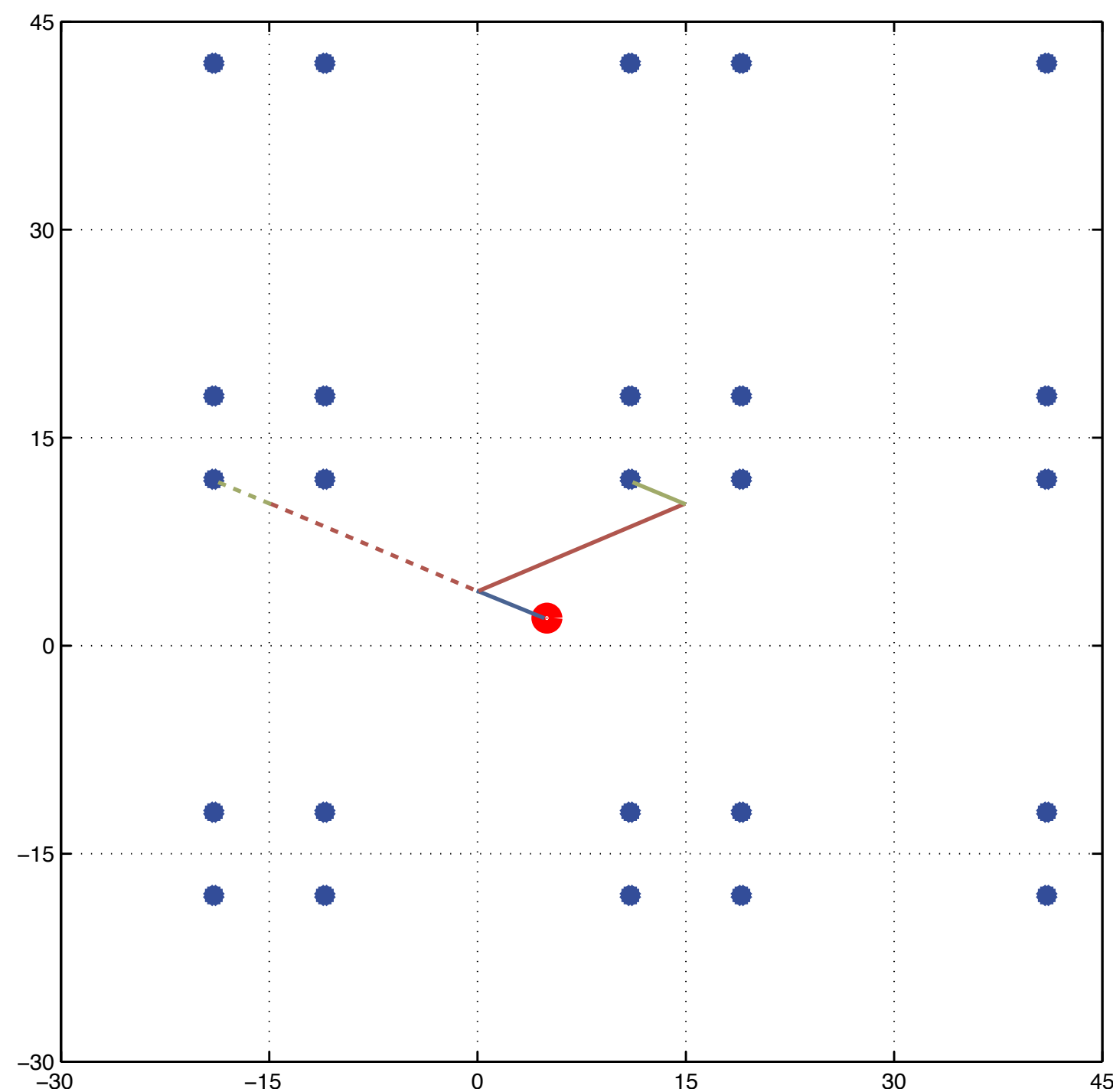
Do that for all walls



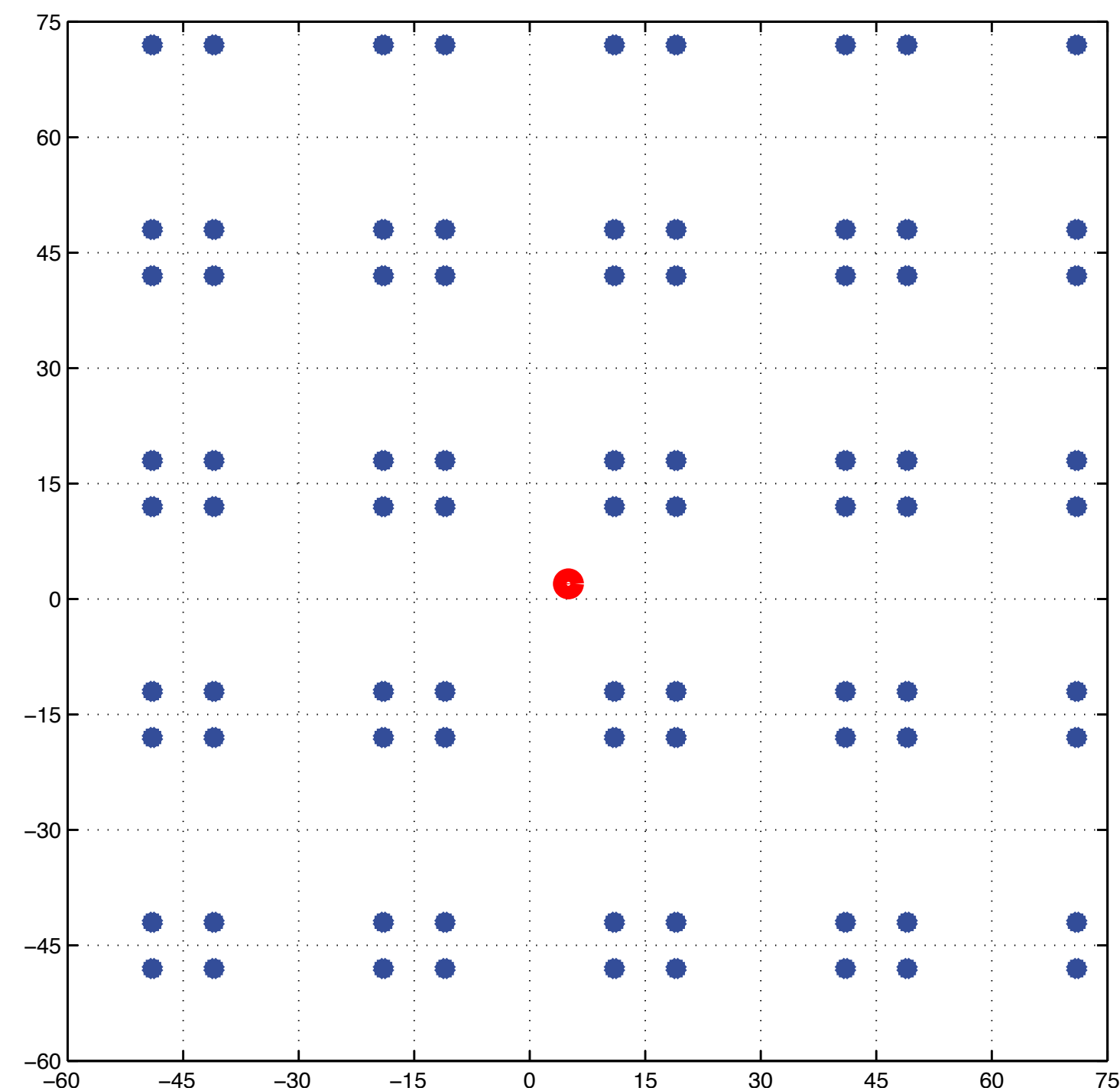
And repeat

- Keep going for as many reflections as you need
 - Note that this becomes exponentially more complex!

Second order reflections



Third order reflections



Blending models

- Image-source model is great for early reflections
 - But gets complicated at the tail
 - Remember 1,000s of echoes per sec!
- Blend image-source model with noise convolution
 - Convolving with exponentially decaying noise can simulate the tail much faster than otherwise

More elaborate models

- Image-source isn't perfect
 - No model for diffusion, problematic with complex room shapes, invalid virtual source problems
- Instead we can use path or beam tracing
 - Randomly follow paths from source to microphone
 - Or beams instead of paths
 - Find hits and use for room model

But there's not just one RIR!

- Each spot inside a room has its own RIR
 - There isn't a single true RIR!
- To really model a room we can:
 - Compute all sound beams explicitly
 - Costly!
 - Blend various RIRs as a source moves in room
 - Easier, but less precise

One room, many RIRs

Compensating for physical rooms

- What happens when you play a simulated room recording inside a room?
 - Not what you wanted!
- We often measure room responses so that we can undo them and present the desired signal
 - Have to both equalize and remove echoes

I Am Sitting in a Room - Alvin Lucier



<https://www.youtube.com/watch?v=bhtO4DsSazc>

Recap

- Reverberation via filtering
 - Simple filters
 - Room Impulse Responses
 - Measuring room responses
- Geometric models of rooms
 - The image-source model

Reference material

- Artificial reverberation
 - https://ccrma.stanford.edu/~jos/pasp/Artificial_Reverberation.html
- Beam tracing
 - <http://www.cs.princeton.edu/~funk/sig98.pdf>
- Image-source model
 - <https://jontalle.web.engr.illinois.edu/Public/AllenBerkley79.pdf>

Next lab

- Designing simple room simulators
- Measuring reverberation